

Lean information management: Understanding and eliminating waste

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Abstract

This paper deals with the development of a new approach for supporting the improvement of information management and the overall information systems infrastructure. In particular, the paper discusses the application of lean thinking to information management; where information management can be considered to involve adding value to information by virtue of how it is organised, visualised and represented; and enabling information (value) to flow to the end-user (customer) through the processes of exchange, sharing and collaboration. The potential benefits of lean thinking are discussed and the fundamental barriers for its application to information management are highlighted. These include the need to characterise the nature of waste and establish the five principles of; value, value streams, flow, pull and continuous improvement in the context of information management. It follows that the core contribution of this paper is the development of an understanding of these critical elements and the creation of a conceptual framework for a set of lean principles within the context of information management. This framework offers a unique and arguably generic approach for supporting the retrospective improvement of information management systems and the overall information systems infrastructure.

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1. Introduction

Information and systems for its management are critical elements for the efficient and effective operation of today's knowledge dependent organisations. From an organisation's perspective the objective of information management is to ensure that valuable information is acquired and exploited to its fullest extent (Willpower Information, 2005). The activities of information management can be considered to involve the creation, representation, organisation, maintenance, visualisation, reuse, sharing, communication and disposal of information (Larson, 2005; Treasury Board of Canada, 2005). At a more conceptual level these elements can be considered to involve adding value to information by virtue of how it is organised, visualised and represented, and enabling information (and hence value) to flow to the end-user through the processes of

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exchange, sharing and collaboration. Furthermore, it is desirable that these elements are performed efficiently with the minimum of waste.

Because of the critical dependency of organisations on information, improving its overall management can yield significant operational benefits to all areas of an organisation and importantly its overall efficiency, competitiveness and responsiveness (Chaffey & Wood, 2004; Dietel, 2000; Moran, 1999). Such improvements generally involve either expanding the amount of information managed or implementing an additional or new information management system. These information management systems or information systems (IS) are generally commercial off-the-shelf (COTS) software applications or suites that are based on standards, languages and processes and include a variety of tools and methods to support the specific activities of information management. In a modern organisation there will be a large number of different systems including finance, payroll, customer relationship management (CRM), product data management (PDM) and inventory management systems. This set of IS or software applications form a complex system which in itself needs to be well aligned to the organisation, efficient and also responsive. Because of this, information management systems that are not well aligned to the organisation or the existing IS infrastructure can have a significant detrimental effect on the organisation and its performance. To alleviate some of these issues there are a variety of methods for determining the information that needs to be managed; a wealth of tools for managing particular types of information; a range of integrated information system suites; and various techniques to support IS planning and the assessment of alignment, fit and user satisfaction; there are however few techniques specifically aimed at improving the efficiency and responsiveness of the overall system or infrastructure.

In contrast to the lack of supportive methods for improving the overall information management system or infrastructure, there are a number of well established techniques that support continuous process improvement of manufacturing and production systems. In general the approaches aim to increase production efficiency (productivity) and quality, and reduce work-in-progress, stock levels and unnecessary product handling. Such improvements can ultimately realise increases in profitability and improve manufacturing flexibility, capability and responsiveness. The most widely adopted approaches include the philosophy of lean thinking (Womack, Jones & Roos, 1990) and the Kaizen (Imai, 1986), single minute exchange of die (SMED) (Shingo, 1985), Six Sigma (Pyzdek, 2003) and design for changeover (DFC) (McIntosh, Culley, Mileham, & Owen, 2001) methods. With the exception of lean thinking these methods can be summarised as frameworks and techniques that focus on improving processes and practices (Kaizen and SMED), product quality (Six Sigma) or changeover performance (DFC). In contrast, lean thinking is a more generic philosophy or framework, and has the potential to be applied to any system or process in order to identify critical areas of improvement and ultimately bring about such improvements. The principles of lean involve eliminating waste and ensuring value flows, which arguably reflect the conceptual model of information management previously proposed. Fundamental to the successful application of lean is the identification of value, understanding of flow and characterisation of waste. In the areas of manufacturing and production the underlying nature of waste is generally well understood and usually visible. However, waste within the context of information management is less clear and not generally as visible. Therefore, prior to the application of lean principles to the processes associated with information management, it is first necessary to develop an understanding of waste within the context of information management and characterise the types of waste present within the overall information management system and infrastructure.

In order to develop an understanding and a theoretical basis for the application of lean thinking within the context of information management this paper discusses the development of a model of waste and the key principles for lean information management. The paper firstly discusses techniques for improving information management and highlights the need for more fundamental approaches that support the overall improvement of the information management system. It then introduces the concept of lean thinking and discusses its evolution and application to processes beyond that of traditional manufacturing. Following this, the underlying nature of waste within the context of information management is characterised and then mapped to the key issues for information management within small to medium-sized organisations. The understanding of waste is then discussed with respect to the traditional model and methods associated with lean manufacturing and the principles of a lean information management strategy are proposed.

2. Improving information management

In this section the current practices and state-of-the-art for improving information management within organisations are discussed. In particular, the review highlights the relative lack of overall principles or frameworks for improving information management *per se* and the overall information management infrastructure including people, practices, processes, systems and the information itself. Rather the majority of the approaches focus on elements of each of these aspects. Including particular sources, specific processes, IS (software applications), integration of the infrastructure, alignment to the organisation and user satisfaction. Whilst these aspects are all important it is arguable that a more holistic or systemic approach has the potential to deliver considerably greater benefits for the organisation. The various areas of information sources, processes (IS), IS integration and assessment of the IS infrastructure are summarised in the following sections.

2.1. *Improving the management of sources of information*

Over the last decade the amount of information that is created, stored and accessed within an organisation has risen exponentially and continues to rise. In fact in 2002, 69% more information was recorded than in 1999, with about 800 MB of recorded information being produced per individual each year (SIMS, 2003). This ever-increasing volume of information is driven by the variety, diversity and numbers of sources, tools and methods for generating information, which are themselves continually rising in number. Because of this increasing volume and diversity of information types (including information records, information sources, electronic files and electronic documents) dedicated systems are developed for archiving and accessing particular types of information (documents/files) including for example records and database management systems (RDBMS), document management systems (DMS) or content management systems (Google, 2006a; Hummingbird, 2006) and product data management (PDM) systems (CAD/CAMNet, 2006; Google, 2006b).

2.2. *Improving information management for business processes*

The second area associated with improving information management focuses on supporting business processes themselves and more specifically the creation and implementation of IS (software applications) for specific business processes or the activities of a particular group within the organisation. Common examples include customer relations management (CRM) tools, accounting and payroll systems, inventory systems and logistics (Microsoft, 2006; Oracle, 2006; SAP, 2006). In addition to these more generic business processes, many domain specific IS have been researched and developed. These include systems for design and manufacture of electronics assembly (Tirpack, 2000), laboratory information in food processing factories (Cagindi & Otles, 2004), industrial environmental assessment (Carlson & Palsson, 2001) and web-based project management (Stewart & Mohamed, 2004).

2.3. *Improving the integration of the IS infrastructure*

The third dimension for improving information management involves the integration of the IS infrastructure. The infrastructure of an organisation generally consists of a large number of IS that support various individuals, groups, departments and processes across the entire organisation, and increasingly multiple sites. This complicated system of interrelated elements ultimately needs to function as a whole in order to best support the organisation. The need for a unified or integrated infrastructure is highlighted in a number of texts (BSI, 2005; Curtis & Cobham, 2002; Laudon & Laudon, 1996). The need for an integrated or unified infrastructure has driven the development of materials resource planning (MRP I) to manufacturing resource planning (MRP II) and enterprise resource planning (ERP) systems that support a large number of core business processes and represent the closest commercially available single unified system (Microsoft, 2006; Oracle, 2006; SAP, 2006).

2.4. Assessing the IS infrastructure

The fourth area of work deals with various aspects of what can be thought of as performance of the IS infrastructure. This includes alignment with an organisation and its core competencies, assessment of fit, user satisfaction and benefits. The subject of assessing and evaluating IS has received some attention (Delone & Van de Ven, 2000; Kanellis, Lycett & Paul, 1999). This work involved post implementation analysis using a number of metrics or key performance indicators (KPIs) which consider aspects such as decision making, innovation, information acquisition and information distribution (Irani & Love, 2001; Kanellis et al., 1999). Such approaches typically result in recommendations for corrective action of a particular implementation or provide a framework which supports the formulation of an IS strategy based on the business strategy and core competencies of the organisation. In addition to this, the challenges surrounding the IS infrastructure for many large organisations has matured to those of effective planning and implementation (Palanisamy, 2005; West & Daigle, 2004) and change management (Edwards & Humphries, 2005). These techniques are important for the development and implementation of more effective IS strategies but do not provide a fundamental set of principles or framework for more effective information management *per se*.

3. Lean thinking

The concept of 'lean' was first introduced by Womack, Jones and Roos (1990) in order to describe the working philosophy and practices of the Japanese vehicle manufacturers and in particular the Toyota Production System (TPS). More specifically, it was observed that the overall philosophy provided a focused approach for continuous process improvement and the targeting of a variety of tools and methods to bring about such improvements. Effectively, the philosophy involves eliminating waste and unnecessary actions and linking all the steps that create value. In 1996 the initial concept of lean was more extensively defined and described by five key principles (Womack & Jones, 1996):

1. Specify value – Define value precisely from the perspective of the end customer in terms of the specific product with specific capabilities offered at a specific time.
2. Identify value streams – Identify the entire value stream for each product or product family and eliminate waste.
3. Make value flow – Make the remaining value creating steps flow.
4. Let the customer pull value – Design and provide what the customer wants only when the customer wants it.
5. Pursue perfection – Strive for perfection by continually removing successive layers of waste as they are uncovered.

In order to introduce lean thinking within manufacturing environments the philosophy relies on the identification and elimination of waste and it is this fundamental aspect, which must first be understood, in order to effectively target and apply the various lean tools. In general lean transformations employ techniques such as Kaizen (Imai, 1986), SMED (Shingo, 1985), Six Sigma (Pyzdek, 2003), value stream mapping (Hines & Rich, 1997) and the five 'S's (Warwood & Knowles, 2004) in order to remove waste and deliver improvements in specific areas. However, it is the fundamental understanding of waste that is critical to successful lean transformation.

3.1. The seven deadly wastes

Within the context of manufacturing systems there exist seven types of waste. These were first identified by Ohno (1988) of Toyota and reported by Womack and Jones (1996). The seven wastes include:

1. Overproduction – Occurs when operations continue after they should have ceased. This results in an excess of products, products being made too early and increased inventory.

2. Waiting – Sometimes referred to as queuing and occurs when there are periods of inactivity in a downstream process because an upstream activity has not delivered on time. Sometimes idle downstream processes are used for activities that either do not add value or result in overproduction.
3. Transport – Unnecessary motion or movement of materials, such as work in progress (WIP) being transported from one operation to another. In general transport should be minimised as it adds time to the process during which no value is added and handling damage can occur.
4. Extra processing – Extra operations such as rework, reprocessing, handling or storage that occur because of defects, overproduction or excess inventory.
5. Inventory – All inventory that is not directly required to fulfil current customer orders. Inventory includes raw materials, work-in-progress and finished goods. Inventory all requires additional handling and space. Its presence can also significantly increase extra processing.
6. Motion – Refers to the extra steps taken by employees and equipment to accommodate inefficient layout, defects, reprocessing, overproduction or excess inventory. Motion takes time and adds no value to the product or service.
7. Defects – Finished goods or services that do not conform to the specification or customer's expectation, thus causing customer dissatisfaction.

In addition to these seven deadly wastes, [Womack and Jones \(1996\)](#) identified an eighth category. This relates to the underutilisation of people and in particular their ideas and creative input for improving the processes and practices. This eighth category of waste is not dealt with in this work as it is arguable that the consequence of this eighth category is inherent in the seven wastes previously defined.

3.2. The application of lean thinking

Over the last decade the principles of lean have been extensively applied to manufacturing operations and production environments in not only the automotive ([Kochan, 1998](#)) and aerospace industries ([Crute, Ward, Brown & Graves, 2003](#)) but increasingly many small to medium-sized manufacturing organisations ([Achanga, Shehab, Roy & Nelder, 2006](#)) and the construction industry ([Pheng & Fang, 2005](#)). The relative success and commercial benefits of lean thinking include a focused enterprise-wide approach to continuous improvement, increased productivity, improved quality and improved management. As a consequence, both academia and industry are beginning to investigate the application of the lean philosophy beyond the primary manufacturing system. This includes what can be thought of as secondary and supportive processes for manufacturing, production processes beyond the traditional manufacturing and physical products, and also other aspects of the business, such as administrative processes ([Innovations, 2005](#)). For example, [Dasari \(2005\)](#) describes the application of lean to the software development process in order to improve quality and build in a continuous improvement cycle. The study defines waste, the principal contributors to value, the process for allowing customers to pull demand and product optimisation. [Haque and Moore \(2004\)](#) investigate the application of lean to the product introduction process and propose the key activities, tools and techniques that constitute a lean product introduction process. In contrast to these process oriented studies, work by [Ranky \(2005\)](#) deals with the design of a lean infrastructure and in particular the control architecture necessary for lean and flexible manufacturing systems.

3.3. Lean thinking and information management

In more general terms it is arguable that the principles of lean thinking and in particular the removal of waste and pursuit of perfection can be applied to any system where product flows to meet the demand of the customer, user or consumer (another system). These elements are certainly true for information management and systems for its management, where information flows and work is undertaken to add value to the information. This value-flow model as applied to a production system is depicted in [Fig. 1](#) which also shows the analogous model of value and flow for information management systems.

It is arguable that the analogous value-flow model for information management can be applied to any information processing activity. This includes for example, the processes of explicit data generation for

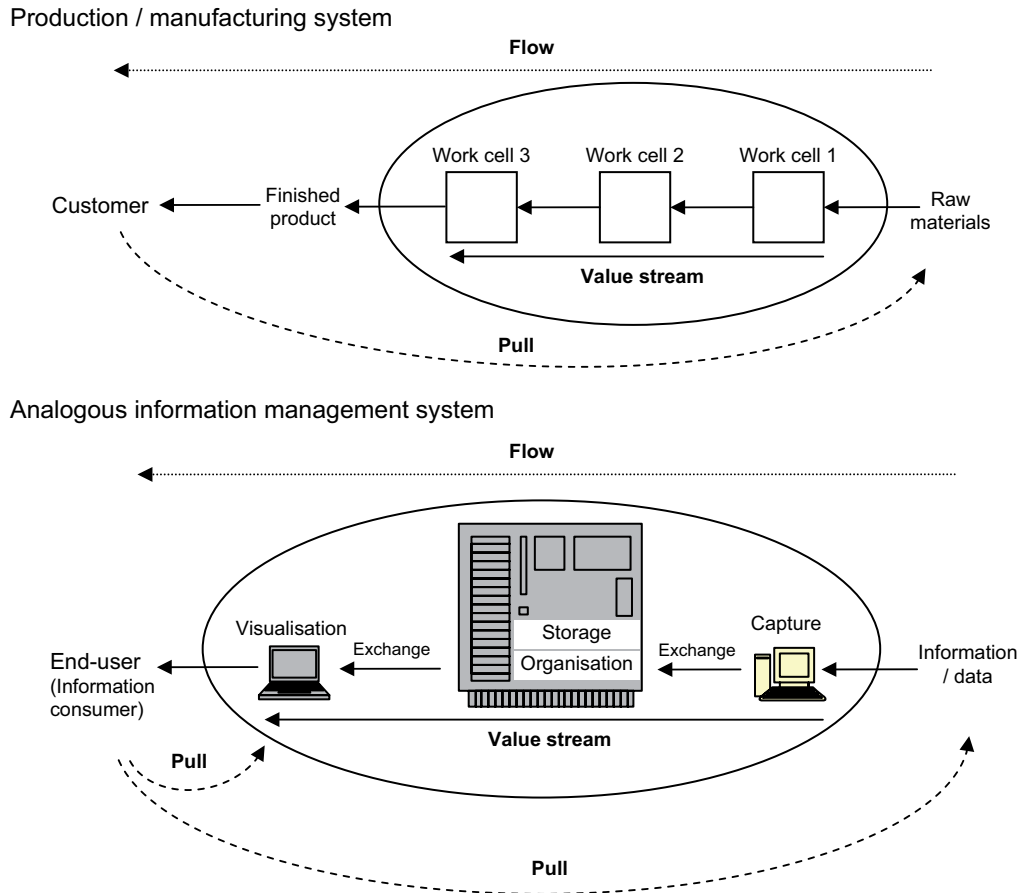


Fig. 1. The value-flow model as applied to information management.

operational decision-making or the acquisition and management of information records for background information or archiving. In both these examples, there is an intrinsic value in the information (data) itself and further value is generated by virtue of the mechanisms by which it is organised, represented, exchanged and visualised. In addition, these mechanisms (or lack of) and the information itself may generate what can be thought of as waste. This waste may include the effort necessary to overcome difficulties in retrieving or accessing information, or the activities required to confirm and correct inaccurate information (e.g. gathering further information or verification).

In general, the notion of waste within the context of information management can be considered to include the additional actions and any inactivity that arise as a consequence of not providing the information consumer immediate access to an adequate amount of appropriate, accurate and up-to-date information. This concept is again analogous to the principles of lean thinking in a manufacturing context. It therefore follows that the philosophy of lean within the context of information management, is to identify and enable focused improvements on the various aspects of information management previously defined, in order to eliminate waste and improve the flow of value. Further, such improvements can, as has been the case in the manufacturing sector, lead to improvements in the efficiency, productivity and quality of the overall process (information management) and product (information). All of which better supports an organisation in undertaking its core activities and sustaining its long-term competitiveness.

As previously stated, in principle, the concept of lean can be applied to any information processing activity and its limitations are only constrained by the ability to identify waste and define value, which are arguably less visible and more subjective. This is in considerable contrast to the manufacturing sector where there exists a culture of performance measurement and identifying and characterising aspects of waste is relatively

straightforward. This is due to the physical and visible nature of the system considered. For example overproduction is very visible and its effect tangible. In contrast to this, where digital systems are considered the various dimensions of waste do not occupy an equivalent space, the effects are less tangible and the value, or lack of, is far less clear and arguably highly subjective. This is particularly the case where activities such as background information and research activities are dealt with. It follows that whilst lean has the potential to support improvements in information management systems a fundamental barrier to its application or introduction is a lack of understanding of waste and also the concept of value.

4. Research method

The focus of the work reported in this paper is on lean thinking and in particular its application to information and the overall system for its management within organisations. More specifically this study focuses on the perspective of small to medium-sized organisations. These are organisations with a workforce of up to 250 employees and, or a turnover of less than £28 million (Europa, 2004). As previously stated, central to successful lean transformation is the need to understand and characterise waste and also value from the customer or end-users perspective. Where information is considered, it is particularly difficult to define the continuum of 'value' for the user of customer. This is because the value proposition is likely to depend upon whether the information supports decisions making, familiarisation or general background and also whether it offers current value or potential value in the future.

In order to overcome this, the underlying assumption is made in this work that the information managed is of some value to the organisation and/or individuals within the organisation and that the IS implemented support the flow of value. The justification for this approach is that many small to medium-sized organisations possess limited resources and only undertake or invest in activities which add some perceived value. Therefore, the focus of this study is more on ensuring value flows and that the customer or in this case the end-users (employees) can pull value successfully. Hence, waste can be considered to be represented by the barriers to improving information flow and the ability of customers (end-users) to pull product and hence value.

What can be thought of as the barriers to information flow within SMEs has been previously reported by the authors in a study of the information management issues facing SMEs (Hicks, Culley & McMahon, 2006). This extensive study involved 10 organisations and identified 18 core issues from an initial 180 issues. It is arguable that these issues represent the fundamental barriers to improvement and are representative of the typical information management issues encountered by small to medium-sized organisations. The issues identified in this previous study are therefore used as the basis for characterising the nature of waste with respect to information management and the IS infrastructures.

5. Characterising waste in the context of information management

As previously stated in Section 4, for the purpose of this study, the causes of waste within the context of information are inferred from a previous study by the authors (Hicks et al., 2006). The findings of this study and in particular the information management issues for small to medium-sized organisations are summarised in Table 1.

Through an evaluation of these issues with respect to the concept of information flow four fundamental causes of waste can be elucidated. These include:

1. Information that cannot flow because it has not been generated, a process is broken, or a critical process is unavailable.
2. Information is unable to flow because it cannot be identified and flow activated or shared processes are incompatible.
3. Excessive information is generated and maintained or excessive information flows, and as a consequence, the most appropriate and accurate information cannot be easily identified.
4. Inaccurate information flows resulting in inappropriate downstream activities, corrective action or verification.

Table 1

The fundamental barriers to improving information management in SMEs

1	Issue	Description and example
1	Information exchange	● Inability to exchange information between information systems. Including: 1. Manufacturing Resource Planning and Product Data Management 2. Manufacturing Resource Planning and Computer Aided Design 3. Manufacturing Resource Planning and Customer Relationship Management 4. Product Data Management and product configurator ● Use of different standards for classifying and organising resources, inventory and products: 1. By organisations 2. Over different sites 3. By departments—e.g. part codes for components
2	Manual systems and data entry	● Due to the inability to exchange data automatically ● Overcome limitations of another systems—such as data input speed ● Manual methods for recording time (timesheets) ● Paper based processes such as works orders
3	Monitoring, control and costing	● Unable to generate accurate up-to-date quotations ● Unable to monitor time, WIP & shop floor processes ● Difficult to undertake variance analysis
4	Information flow from customers and/or sales	● Poor information provided by sales to technical departments ● Lack of formalised sales process and as a result customers fail to supply appropriate information
5	Functionality of IS	● Inability of Manufacturing Resource Planning system to manage cost and contact data ● Inability of Product Data Management system to handle both 2D and 3D drawings files and files from different Computer Aided Engineering systems
6	Information storage	● Cost of keeping legacy data ● Increased amount of information to manage as a result of ISO 9000/2
7	End-user developed applications over COTS information systems	● Overcome limitations of commercial-off-the-shelf software applications ● Avoid expensive customisation of commercial-off-the-shelf applications ● Created to supplement information systems. Including: 1. Manufacturing Resource Planning 2. Product Data Management 3. Sales 4. Customer Relationship Management
8	Information systems use and maintenance	● Long timescales necessary to fully integrate Manufacturing Resource Planning system ● Cost of training both in terms of finance and time ● Wide spread use of ‘Gatekeepers’—large knowledge gap if gatekeepers leave. Gatekeepers’ are experienced or highly trained users—often only one or two for a given system.
9	Numbering and traceability of machines, assemblies and parts	● Transition from a numbering system that supports 2D CAD modelling to 3D modelling has created two parallel systems ● Difficult to tracing/track design changes made during commissioning which means that actual build is different to recorded build

Table 1 (continued)

1	Issue	Description and example
10	Information availability and accessibility	● Sales require information from technical departments and service engineers but are generally unable to access the systems ● Difficulties locating electronic files (word, pdf, excel etc.) on personal computers and file stores—need a centralised approach for the management of electronic files
11	Information systems implementation and customisation	● Time and resources necessary to fully implement and align Manufacturing Resource Planning system to organisation
12	Information identification, location and organisation	● Identifying and locating files on PCs and servers ● Backup issues—which files and where? ● Migration issues—which files to keep? ● Difficult to locate information for production of e-manuals
13	Information completeness and accuracy	● Partially entered information in information systems ● Errors during input or transfer particularly if manual or semi-automated
14	Implementation and operation of quality systems	● Resources necessary to implement and manage a quality systems ● Large amounts of information produced as a result of achieving and maintaining certification
15	Information duplication	● Multiple instances of information make it difficult to identify the most accurate/up-to-date ● Frequently hold additional instances in Manufacturing Resource Planning system and other information systems. For example organisations often maintain a Bill Of Materials in the Product Data Management system and also the MRP system
16	Information currency	● Old information on product range, revisions and options ● Old suppliers literature ● Out-of-date Bills Of Materials
17	Paper systems over COTS information systems	● Use of paper based Master Records for drawings, parts, inventory and machine numbers
18	Information systems strategy and planning	● Strategy and planning driven by accounting and finance and/or the directors responsible for the financial stewardship of the company and not well aligned to the overall organisation

These four causes of waste give rise to four corresponding types of waste, defined below.

1. *Failure demand*: This includes the resources and activities that are necessary to overcome a lack of information. This may include generating new information and/or acquiring additional information.
2. *Flow demand*: This concerns the time and resources spent trying to identify the information elements that need to flow.
3. *Flow excess*: This relates to the time and the resources that are necessary to overcome excessive information i.e. information overload (Edmunds & Morris, 2000; Fortune, 1994).
4. *Flawed flow*: This includes the resources and activities that are necessary to correct or verify information. It also includes the unnecessary or inappropriate activities that result from its use.

To provide further insight into the four types of waste, the issues presented in Table 1 are classified with respect to the categories of waste in Table 2. Some of the issues in Table 2 are related and through

Table 2

A classification of information management issues with respect to waste

	Issue	Implications for waste	Waste category
1	Information exchange	Inability to automatically exchange information and enable value to flow results in additional processes necessary to overcome this lack of functionality/poorly functioning process	Flow demand
2	Manual systems and data entry	Processes and resources necessary to overcome (1) and also where processes are unavailable	Failure demand
3	Monitoring, control and costing	Information is required but has not been generated and cannot flow	Flow demand
4	Information flow from customers and/or sales	Information does not flow and processes are not well defined so additional effort is necessary to acquire information	Failure demand and flow demand
5	Functionality of information systems	Inability to perform certain functions to support the management and flow of information requires additional resources	Failure demand
6	Information storage	Excessive information is stored—partly because of a lack of understanding of potential value	Flow excess
9	Numbering and traceability of machines, assemblies and parts	Additional effort required to locate up-to-date/accurate build	Flow demand
10	Information availability and accessibility	Time and effort necessary to identify information to flow and unavailability of processes to support flow	Flow demand and failure demand
12	Information identification, location and organisation	Resources necessary to identify the information to flow. The level of resources increases as the amount of information increases	Flow demand and flow excess
13	Information completeness and accuracy	Effort necessary to verify and/or correct information and the effect of its use	Flawed flow
15	Information duplication	Effort necessary to arbitrate between multiple instances of information	Flow demand and flawed flow
16	Information currency	Out-of-date or inaccurate information requires additional effort to verify and/or update information	Flawed flow
7	End-user developed applications over COTS information systems	Acquisition of additional resources to overcome existing process limitations and also in an attempt to minimise cost (waste)	Failure demand and waste reduction
17	Paper systems over COTS information systems	Master records are maintained in an effort to reduce waste however they may arise due to failure of existing processes	Waste reduction and failure demand
8	Information systems use and maintenance	Activities are perceived as a waste by members of the organisation but arguably undertaking these activities might improve flow and eliminate waste	Value
11	Information systems implementation and customisation	Activities are perceived as a waste by members of the organisation but arguably undertaking these activities might improve flow and eliminate waste	Value
14	Implementation and operation of quality systems	Activities are perceived as a waste by members of the organisation because the value to the organisation is not understood or insufficient	Value
18	Information systems strategy and planning	A lack of understanding of the value of information and its flow across the organisation can result in a poorly performing system and arguably waste	Value and waste

consideration of the causality it is possible to reduce the 18 issues to nine fundamental issues (Hicks et al., 2006). However, for the purpose of examining waste all the 18 issues are dealt with as these provide a more detailed understanding of the possible causes of waste. The consequences of this are that where issues are related their exclusive classification against a single dimension of waste is difficult. Notwithstanding this, issues are classified with respect to their primary type of waste: flow demand, flow excess, flawed flow or failure demand.

It is clear from Table 2 that not all of the 18 issues can be classified with respect to waste. In total, 12 issues can be directly classified with respect to waste, two issues can be partially classified and the remaining four issues relate to the concept of value. For the two issues that can be partially classified; the *use of paper based systems over commercial-off-the-shelf (COTS) software applications and end-user developed applications over COTS applications*, these can be considered to arise because the commercial-off-the-shelf applications fail to adequately perform certain processes (failure demand) and also as a result of an attempt to reduce waste and more specifically cost (implementation and customisation of software applications) and time (employees).

Of the four issues that cannot be directly classified with respect to the concept of waste, three involve IS implementation, use and maintenance. For all three of these issues the perception of the organisation or certainly a large number of employees is that the resources and effort associated with fully implementing and integrating these systems is a relative waste. This is presumably because the employees and perhaps the organisations do not believe the systems offer sufficient value or at least they do not understand the potential value to the organisation. In many cases this value may well be realised through the elimination of waste as proposed in this work. In the case of the last issue; IS strategy and planning, it is arguable that this issue arises because of a lack of understanding of the concepts of waste and value as applied to information and also contributes significantly to the potential level of overall waste within the information management system.

5.1. A comparison to the seven types of waste of manufacturing systems

In this section the four categories of waste are compared and contrasted with the seven traditional wastes associated with manufacturing systems, presented in Section 3. A comparison of the classifications reveals that the concepts of flow excess, flow demand, failure demand and flawed flow can be mapped directly to the traditional model and in particular overproduction, waiting, extra processing and defects, respectively. This is shown in the upper portion of Table 3. In contrast, for the purpose of information management there are no analogous dimensions of waste for transport, inventory or motion. The primary reason for this is because of the electronic or digital nature of the system being considered. In general, events (data exchange) within the system appear to occur almost instantly and capacity (data transfer) can be increased at relatively minimal cost in order to meet demand. The author notes that in exceptional cases this is not always true; however, such large scale bespoke systems are not dealt with in this work. In a similar manner to this, where inventory is considered within a digital system, the physical space occupied by stock or work in progress and its relative cost are, in general, considered to be trivial.

Notwithstanding this, it is possible to draw analogies between the categories of transport, inventory and waste in the traditional manufacturing model and the information model if the information users are also considered. For the purpose of this work, these analogies are drawn from general observations of the IS, user behaviour and the overall strategies adopted within the organisations studied. In the case of transport, it is arguable that from a lean perspective unnecessary movement of materials frequently occurs with mass correspondence or e-mails. Whilst there may be negligible waste associated with the movement of the material itself, the waste manifests itself in the effort consumed by the recipients in determining whether or not the

Table 3
The seven deadly wastes of manufacturing and information management

	Domain		
	Manufacturing systems	Information management	Information users
1	Overproduction	Flow excess	
2	Waiting	Flow demand	
3	Extra processing	Failure demand	
4	Defects	Flawed flow	
5	Transport		Mass electronic communication
6	Inventory		Legacy databases and file archives
7	Motion		Gatekeepers/single seat licences

'pushed' (Campbell, Culley & McMahon, 2005) information offers any value. Similarly, in the case of inventory, many users maintain an ever-increasing archive of legacy information which whilst it may not incur significant financial cost it does affect the performance of the user in retrieving information. Finally, in the case of motion, it was observed that many organisations adopted a 'Gatekeeper' and/or single seat licence philosophy. This is where one individual is trained to use a particular software application. Only a single or small number of users are trained in an attempt to reduce both training and licence costs. However, the consequences are that the organisation depends upon the gatekeeper(s) and frequently particular workstations (i.e. licensed computers) for particular information. This often results in users waiting for and moving between computers, which therefore creates unnecessary motion and also waiting.

6. The principles of lean information management

This section develops the key principles of a lean approach for information management. This approach is applicable to information *per se* and the overall systems for its management. In particular, the five key principles identified by Womack and Jones (1996) and discussed in Section 3 are elucidated with respect to information management. These principles are summarised in Table 4 and discussed in the following sections.

6.1. Value

In general, information and IS support the activities of end-users and, in particular, their ability to undertake work, where work may be defined as those activities that add value to either the product or service delivered by an organisation and ultimately the customer. From an information perspective, it is also important that the information and the functionality embodied in the systems for managing information supply 'value' to the information consumer. For the purpose of this work, the end-users may also be considered to be a 'customer' of the IS and hence demand value. The term information consumer is used to differentiate between the ultimate customer of the organisation and the customer of the IS. Whilst this work does not consider the subject of information value in detail it is arguable that organisations manage three classes of information: information that has to be managed due to regulatory and statutory requirements, information that is necessary for the effective management or operation of the organisation and information that supports core competencies and more specifically the processes that support the organisation in delivering their product(s) or service(s). Furthermore, within each class there may be a variety of explicit information (data), background information and legacy information. Each of these classes and types may represent a different value proposition to the organisation. As a consequence, identifying and defining the value may require the involvement of a wide variety of stakeholders, detailed audits and assessment of information needs, and in-depth discussions, particularly where the concept of value is more subjective. There are a variety of tools that can support these activities including information audits (Buchanan & Gibb, 1998), information needs analysis (Nicholas, 2000).

6.2. Value stream

For the purpose of information management the value stream can be considered to represent the series of processes and activities that ultimately result in the presentation of the information to the information consumer. This includes the capture, representation, exchange, organisation, retrieval and visualisation of information. In reality it is not unusual for the overall process to involve a number of actors and IS. For the purpose of reducing waste it is important that the sequence (network) of processes are well integrated and where possible events or activities are automated in their operation. Furthermore, it is important that all information consumers (users of the IS) understand the different value propositions and also what can be thought of as the direct and indirect value to the organisation. Where direct value may benefit the particular consumer directly and indirect value may benefit another department or consumer group but require that one or more consumers (actors) perform a specific task or procedure to enable the value to flow. In the area of value stream mapping (VSM) there are a variety of tools, discussed by Hines and Rich (1997), which can be applied to information processes. Furthermore, techniques such as information flow mapping (AMT, 2006)

Table 4
The key principles of a strategy for lean information management

Principle	Description
Value	Information and functionality must supply value to the customer (end user) 1. Only manage valuable information—information that has to be managed and information that supports the core business activities 2. Remember users benefit from the system and only use it if it offers direct value or they understand the indirect value i.e. for another department
Value stream	1. Ensure that the series of processes and activities that deliver information are mapped. This includes processes that support the capture, representation, exchange, organisation, retrieval and visualisation of information 2. Ensure that the sequence (network) of processes that support information management are integrated
Flow	1. Information should be available real time—as soon as it is generated/acquired 2. Ensure that all information processes and support processes occur in the shortest possible time 3. Procedures and processes should be invoked and performed in the simplest way possible 4. Minimise duplication of information within the organisation, across departments, and customers and suppliers 5. Minimise amount of out-of-date or unnecessary information within the organisation, across departments, and customers and suppliers 6. Minimise duplication of effort within the organisation, across departments and customers and suppliers
Pull	1. Information and additional functionality should only be delivered as it is demanded by the users (customers) 2. To facilitate pull, the interfaces, methodology and procedures need to be consistent across the organisation 3. Minimise the dependency on IT staff and programmers for implementation. Support users to undertake local customisation and promote ownership—end-user developed systems
Continuous improvement	1. Regularly review infrastructure and processes—remember information systems, business processes, and processes that support products and services change and as such opportunities for improvement will also change 2. Support rapid implementation and training. This is the time from system acquisition to full implementation and integration with the business processes

are essential for identifying and understanding all information flow (value streams) and the wide variety of information processes and actors.

6.3. Flow

The ‘flow’ principle of lean suggests that the value creation steps (the value stream) should be made to flow. For information handling activities it is rare that there is ever single piece flow, as there might be in a traditional production environment. In the case of information, the aim is to ensure that information flows efficiently and that the most valuable (appropriate, accurate and up-to-date) information flows. To achieve this information should be available real-time as soon as it is generated or acquired. One of the critical elements in achieving ‘flow’ is arguably the development of an integrated IS infrastructure where data exchange occurs automatically and in realtime. In addition to this, all information processing and support processes should occur in the shortest possible time and all procedures and processes should be invoked and performed in the simplest way possible. It is also important to minimise the duplication of information within the organisation, across departments, customers and suppliers; the amount of out-of-date or unnecessary information within the organisation, across departments, customers and suppliers; and the duplication of

effort within the organisation, across departments and customers and suppliers. Central to achieving this, is the reengineering and optimisation of the overall information process(es) including the roles and activities of all actors and systems.

6.4. *Pull*

The ‘pull’ principle of lean states that manufacturers should only design and provide what the customer wants and only when the customer wants it. Therefore, where information and systems for its management are considered, information and additional functionality should only be implemented if and when it is demanded by the end-user (information consumer). In addition to this, in order to facilitate ‘pull’ by the customer (end-user) the interfaces, methodology and procedures for pulling value need to be consistent across all IS within the organisation. Critical elements in achieving this, are undertaking user/organisational requirements capture (Aurum & Wohlin, 1998), establishing common interface designs (Galitz, 2006) and effective information design (Jacobson, 2000).

6.5. *Continuous improvement*

The fifth principle, continuous process improvement, involves regular reviews of the information management system, all associated infrastructure and processes. Regular reviews are necessary as IS, legislation, business processes, processes that support the delivery of products and services change with time, and as such, opportunities for improvement and levels of waste are also likely to change. In addition to the regular reviews, it is also important to support the rapid implementation and training where new or altered IS or processes are implemented. This is the time from system acquisition to full implementation and integration with the business processes. It has been observed in this study, that users often perceive that the implementation activities of new or altered systems and processes are a waste of time, particularly if they do not offer any direct value. Ultimately, the extended implementation time can generate far more waste than the time cost for the user and it is therefore important to minimise this time.

6.6. *Discussion*

The work reported in this paper develops the principles of lean information management and defines the categories of waste within the context of information, systems for its management and the information consumer. These aspects are critical elements for the introduction of lean principles to information based activities but do not explicitly represent a strategy for improving information management. Rather the principles provide a generic framework, which supports and directs an improvement programme and philosophy. It is envisaged by the author that the lean information management principles and framework would form an important part of IS planning; identifying areas for improvement and requirements, as well as supporting the information officer; helping to identify valuable information assets. In order to undertake a lean transformation, a variety of tools and methods are required to investigate specific areas of improvement. Examples of such tools are presented in the preceding sections and are not meant to be exhaustive or stated as being the most appropriate. In fact, it is likely that considerable further work is necessary to both identify appropriate tools and also develop new tools for assessing aspects such as information value. This is particularly the case for information activities that do not involve explicit limited data and are more focused on knowledge creation.

In addition, within the manufacturing sector, the philosophy of lean is well developed and the research community is now focused on issues concerning the role of management and communication in lean implementation (Worley & Doolen, 2006), key performance measures and critical success factors (Achanga et al., 2006), and assessing lean performance (Womack, 2006). It is therefore likely that such issues will also require the attention of researchers from the field of information science,

if lean thinking is to be successfully introduced and its potential benefits maximised within the context of information management.

7. Conclusions

This paper deals with the development of a new approach for supporting the improvement of information management and in particular the performance of the overall information management system or infrastructure. It is argued that there are many tools and methods for improving particular aspects of information management but few general methods or principles that can be applied to information management and its range of activities *per se*. In contrast to this, there are a number of generic techniques or principles that can be applied to manufacturing and production systems in order to bring about improvements. More specifically, the principles of lean thinking have revolutionised manufacturing and production operations and have arguably contributed to the improvement programmes of many of today's leading manufacturing organisations. It follows that the fundamental proposition of this paper is that lean thinking can also be applied to information management; where information management can be considered to involve adding value to information by virtue of how it is organised, visualised and represented; and enabling information (and hence value) to flow to the end-user (customer or information consumer) through the processes of exchange, sharing and collaboration.

The potential benefits and extent of application of lean principles are discussed, and the fundamental barriers for its application to information management are highlighted. These include the need to characterise the nature of waste and establish the five principles of; value, value streams, flow, pull and continuous improvement in the context of information management. In order to characterise waste, the findings from a previously reported study (Hicks et al., 2006) are used as the basis for elaborating the various types of waste and their causes. This revealed four types of waste within the context of information management: failure demand, flow demand, flow excess and flawed flow. The four types of waste are illustrated with examples through their classification to a set of 18 information management issues. This process also highlights the need for a deeper understanding of the concept of 'information value' which is not explicitly dealt with in this paper. The four categories of waste are discussed with respect to the traditional model of seven manufacturing wastes, which reveals that the four categories of failure demand, flow demand, flow excess and flawed flow can be mapped directly to the traditional wastes of extra processing, waiting, overproduction and defects, respectively. However, three of the traditional types of waste; transport, inventory and motion, are not clearly identifiable when considering the digital/electronic system. This is due to the almost instantaneous data exchange and the relatively trivial cost of electronic storage. Notwithstanding this, it is shown that whilst these three types of waste cannot be mapped to the digital system they are still present in the digital-physical system and in particular the behaviour of the end-users.

Using the characterisation of waste and the understanding gained from the study, a conceptual framework for a set of lean principles within the context of information management is proposed. These principles include the traditional elements of value, value stream, flow, pull and continuous improvement. It is argued that the framework and the understanding of waste offer a unique and generic approach for supporting the retrospective improvement of processes and systems that manage information. In particular, the principles aim to support a continuous programme of improvement concerned with the efficiency, quality and overall management of the processes. It is further argued that, as has been the case for manufacturing systems, such improvements are likely to have wider organisational benefits. These may include a greater capability to undertake core activities, improved responsiveness and increased competitiveness.

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